XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

Department of Electrical and Electronic Engineering

Xiamen University Malaysia

Quiz 2

COM105 Engineering Physics (I)

4 Questions

Total Marks (100)

31 May 2022 (Tuesday)

9:00 – 10:15 am (One hour fifteen minutes)

INSTRUCTIONS TO CANDIDATES

1. Get your papers and pen ready, all corresponding working steps and answers **MUST be handwritten**. Make sure you have your data cable or have setup any cloud storage service app in your smartphone and PC (iCloud, Dropbox, Onedrive), which allows you to access your phone's photos or scanned images immediately from your PC.
2. This quiz consists of 4 main questions:

Q1:	5	sub-questions	Q2:	5	sub-questions
Q3:	1	sub-questions	Q4:	2	sub-questions

3. Please follow the requirements of each question, write your **ID, signature, and date** on the top right of **every** pages**.** Indicate the question number i.e., Q1a).
4. Scan your written answer papers with any phone scanner app of your choice, or take picture with the camera app and paste the images to an empty Word document. Make sure the images are clear and readable.
5. Compile all the images and save as **pdf** file. Name the pdf file as **COM105Quiz1_studentID.pdf**.
6. If the file size exceeded 20 MB limit, you may compile multiple **pdf** files according to the question number, as indicated in the file name i.e., **COM105Quiz1_studentID_Q1.pdf**, **COM105Quiz1_studentID_Q2&Q3.pdf**, etc. You are allowed to submit up to 4 **pdf** files in the Moodle Quiz submission tab.
7. You have extra 15 minutes to compile your answer sheet. Submission due at 10:30 am, after which will be considered late and late penalty will be applied.

Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you will be penalized accordingly.

Question 1 (25 marks)

An elevator ascends from rest and reaches its maximum speed of 1.30 m/s in 0.700 s. After which it travels with this constant speed for the next 8.00 s, before slowing down to rest again in 1.0 s. The elevator then descends and reaches the same maximum speed at 1.50 s. If a 48.0-kg woman stands on a spring scale in the elevator, calculate the spring scale register in Newton:

(a) before the elevator starts to move. **(5 marks)**

(b) during the first 0.700 s. **(5 marks)**

(c) while the elevator is traveling at constant speed. **(5 marks)**

(d) during the time it is slowing down while ascending. **(5 marks)**

(e) during the last 1.50 s. **(5 marks)**

Question 2 (25 marks)

A 12.0 kg wooden box is pulled up on a rough incline at a 24.0° angle to the horizontal for a distance of 6.00 m. Given that the initial speed of the wooden box is 1.35 m/s, the pulling force is 10*x* N parallel to the incline, and the coefficient of kinetic friction between the wooden box and inclined surface is 0.410 (*x is the last digit of your ID, i.e., EEE2109308, pulling force = 108 N*). Determine the following:

(a) the work done by the gravitational force on the wooden box. **(6 marks)**

(b) the increase in internal energy of the wooden box–incline system due to friction. **(6 marks)**

(c) the work done by the pulling force on the wooden box. **(3 marks)**

(d) the change in kinetic energy of the wooden box. **(4 marks)**

(e) the speed of the wooden box after being pulled 6.00 m. **(6 marks)**

Question 3 (25 marks)

A weightless wire is wrapped around the rim of a wheel which is free to rotate without friction about a stationary horizontal axis that passes through the centre of the wheel. A mass of 3.80 kg is attached on the free end of the wire. Initially from rest, the mass is released and descends with constant acceleration. If the mass moves downward a distance of 3.60 m in 2.20 s, determine the mass of the wheel. Given that the radius of the wheel is 0.300 m. **(25 marks)**

Question 4 (25 marks)

A round and flat turntable is initially rotating at 3.*x*0 rad/s about a vertical axis through its centre (*x is the last digit of your ID, i.e., EEE210930**8**, initial rotation = 3.**8**0 rad/s*). It has a radius of 2.10 m and a total mass of 130 kg. When a 65.0-kg mass soft-landed on the turntable at a point near the outer edge, calculate the following:

(a) the angular speed of the turntable after the mass lands (treat the mass as a particle). **(15 marks)**

(b) the kinetic energy of the system before and after the mass lands, are the kinetic energies equal? Why? **(10 marks)**